

Rovauto Garage Flow

Current backend/API flow: geocoding, price ranges, garage broadcast, wallet deduction, in-app notifications, handover OTP, delivery acceptance, and admin pricing.

Current Scope

Backend APIs are ready. Real WhatsApp provider envs can stay empty for testing; backend logs the outgoing WhatsApp-style message and accept link. Frontend pages still need to consume these APIs.

Core Rules

Garage search radius defaults to 15km. Search expires after 120 seconds. Handover OTP is sent only in in-app customer notification, not WhatsApp. Customer must accept delivery before the booking appears in service history.

1. Complete Customer Flow

1	Customer enters manual address/city or GPS location. For manual address, frontend calls /locations/geocode to get latitude and longitude.
2	Customer selects vehicle and services. Backend estimates bill as a range from admin city/service/vehicle pricing, or fallback current price + Rs.500.
3	Customer pays Rovauto handling/platform fee through Cashfree. The service bill remains an estimate range, not a fixed final bill.
4	After payment verification, booking becomes SEARCHING_GARAGE and searchExpiresAt is set from GARAGE_SEARCH_TIMEOUT_SECONDS.
5	Backend finds active verified garages within GARAGE_BROADCAST_RADIUS_KM that offer all selected services.
6	Each eligible garage gets a broadcast request and WhatsApp-style message/log with service list, vehicle details, customer area, and accept link.
7	If no garage accepts within 2 minutes, booking and pending requests expire and customer gets an in-app please-try-again notification.
8	If garage accepts, customer gets in-app notification that garage.name accepted, with handover OTP.
9	Customer shares OTP only at vehicle handover. Garage verifies OTP on website; booking becomes IN_PROGRESS.
10	Garage marks delivered; customer gets delivered notification. Customer accepts delivery; booking becomes COMPLETED and appears in service history.

2. Garage Side Flow

Step	Actor	What Happens
1	Garage owner	Submits basic partner application. No images required during application.
2	Admin	Approves, denies, or requests changes. Emails are sent for deny/change/approval.
3	Backend	On approval, creates garage owner, garage, and wallet. Garage is verified but inactive.
4	Garage owner	Recharges at least Rs.1000. Backend credits wallet only after Cashfree verification.
5	Backend	Garage becomes active when verified and wallet balance is at least Rs.1000.
6	Garage owner	Receives booking request log/message and opens accept link on website.
7	Backend	On accept, deducts platform fee from garage wallet before confirming booking.
8	Garage owner	Gets customer location link. Later verifies handover OTP, completes service, and marks delivered.

Garage Wallet Deduction

Rs.300-999 -> Rs.30, Rs.1000-4999 -> Rs.99, Rs.5000-19999 -> Rs.249, Rs.20000+ -> Rs.500, SOS -> Rs.50. If wallet balance is insufficient, accept fails.

3. Admin Pricing

Field	Purpose	Notes
city	City-specific pricing	Used when booking location.city is provided.
serviceId	Service-specific range	Required for every admin price range.
vehicleBrand	Optional vehicle match	More specific ranges win over generic ranges.
vehicleModel	Optional model match	Highest match score is used.
fuelType	Optional fuel match	PETROL, DIESEL, ELECTRIC, HYBRID, CNG, OTHER.
minPrice/maxPrice	Customer estimate range	Service bill is shown as range, not final fixed bill.

Fallback Pricing

If no admin range matches, backend uses current/base service price as lower range and lower + SERVICE_PRICE_RANGE_DELTA as upper range. Default delta is Rs.500.

4. API Routes

Purpose	Route	Access
Geocode manual address	GET /api/v1/locations/geocode	Customer
Create booking	POST /api/v1/bookings/checkout	Customer
Accept delivery	POST /api/v1/bookings/:id/accept-delivery	Customer
Service history	GET /api/v1/bookings/service-history	Customer
Garage requests	GET /api/v1/garage/requests	Garage/Admin
Accept request	POST /api/v1/garage/requests/:requestId/accept	Garage/Admin
Verify handover OTP	POST /api/v1/garage/requests/:requestId/verify-handover-otp	Garage/Admin
Mark delivered	POST /api/v1/garage/requests/:requestId/mark-delivered	Garage/Admin
Garage wallet recharge	POST /api/v1/garage/wallet/recharge/order	Garage/Admin
Verify wallet recharge	POST /api/v1/garage/wallet/recharge/verify	Garage/Admin
Admin price ranges	GET/POST/PATCH/DELETE /api/v1/admin/city-service-price-ranges	Admin
Garage applications	POST /api/v1/garage/applications	Public

5. Notifications

Trigger	Recipient	Message/Result
Garage accepts	Customer	In-app notification: garage.name accepted your service request. Includes handover OTP.
No garage in 2 minutes	Customer	In-app notification asking customer to try again because garages are busy.
Garage marks delivered	Customer	In-app notification asking customer to review and accept delivery.
Broadcast request	Garage	WhatsApp provider call if configured; otherwise server log contains message and accept link.

6. Environment

Env	Default	Purpose
GARAGE_BROADCAST_RADIUS_KM	15	Nearby garage search radius.
GARAGE_SEARCH_TIMEOUT_SECONDS	120	How long booking stays SEARCHING_GARAGE before expiry.
HANDOVER_OTP_TTL_MINUTES	30	How long vehicle handover OTP remains valid.
SERVICE_PRICE_RANGE_DELTA	500	Fallback upper estimate delta.
GARAGE_REQUEST_ACCEPT_PATH	/garage/requests	Frontend path included in garage accept links.
WHATSAPP_PROVIDER_URL/TOKEN/SENDER_ID	empty	Empty means log messages only; fill when real provider is ready.
GEOCODER_PROVIDER	nominatim	Manual address geocoding provider.
NOMINATIM_USER_AGENT	Rovauto/1.0 (...)	Required app/contact identifier for OpenStreetMap Nominatim.

Distance Calculation

Backend stores garage latitude/longitude and booking customer latitude/longitude. It calculates straight-line kilometers using the Haversine formula and includes garages whose distanceKm is <= configured radius.

7. Current Ready vs Later

Area	Status	Notes
Backend APIs	Ready	Routes exist for current requested backend flow.
Frontend accept page	Later	Must call garage request accept/OTP/delivery endpoints.
Real WhatsApp sending	Later	Provider envs needed. Empty envs log messages for testing.
Final service bill/payment	Later	Current system stores estimated range and completion; final bill flow is not implemented.
Admin UI	Later	Backend endpoints exist for city price ranges and garage applications.

Generated for the current Rovauto backend state. Update this document whenever the frontend garage dashboard or final billing/payment flow is implemented.